Investigating the Ability of PINNs To Solve Burgers' PDE Near Finite-Time Blow Up

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NEURAL INFORMATION PROCESSING SYSTEMS

Introduction

Physics Informed Neural Networks (PINNs) have been achieving ever newer feats of solving complicated PDEs numerically while offering an attractive trade-off between accuracy and speed of inference. In this work, we investigate the stability of PINNs near finite-time blow-ups from a rigorous theoretical viewpoint.

Why Finite-Time Blow-Ups ?

- Existing studies with PINNs (including studies of its failure modes) focus on cases where the true solution is "nice".
- An interesting possibility with various differential equations representing the time dynamics of some systems is that their solution might have a finite-time blow-up – a phenomenon where the solution $\boldsymbol{u}$ becomes infinite at some points as t approaches a certain time $T < \infty$, while the solution is well-defined for all $0 < t < T$, $\sup_{x \in D} |\boldsymbol{u}(\boldsymbol{x}, t)| \to \infty \quad as \quad t \to T^-$
- For eg. $\frac{du}{dt} = u^2,\ u(0) = u_0,\ u_0 > 0$ is an ODE s.t it's easy to see that its solution blows-up at $t = \frac{1}{u_0}$
- In certain recent experimental studies with PINNs (Y. Wang et al. 2022), thought provoking experimental evidence was given that PINNs could potentially discover PDE solutions with blow-up even when their explicit descriptions are not known.

(d+1)-Burgers' PDE

We analyse the following PDE in the domain $D \subset \mathbb{R}^d$ and $t \in [-\frac{1}{\sqrt{2}} + \delta, \delta]$

$$\partial_t \boldsymbol{u} + (\boldsymbol{u} \cdot \nabla)\boldsymbol{u} = 0\ ; \quad \boldsymbol{u}(t = t_0) = \boldsymbol{u}_{t_0},\ t_0 = -\frac{1}{\sqrt{2}} + \delta$$

Then for the neural net $\boldsymbol{u}_\theta(\boldsymbol{x}, t) \in \mathbb{R}^d$, we define the residuals as

$$\mathcal{R}_{\text{pde}} := \partial_t \boldsymbol{u}_\theta + (\boldsymbol{u}_\theta \cdot \nabla)\boldsymbol{u}_\theta\ ; \quad \mathcal{R}_{\text{t}} := \boldsymbol{u}_\theta(t = t_0) - \boldsymbol{u}(t = t_0)$$

The L^2 population risk of the predictor that we are interested in is,

$$\int_\Omega \|\boldsymbol{u}(\boldsymbol{x}, t) - \boldsymbol{u}_\theta(\boldsymbol{x}, t)\|_2^2 \, \mathrm{d}\boldsymbol{x}\, \mathrm{d}t$$

where $\boldsymbol{u}$ is the exact solution and $\boldsymbol{u}_\theta$ is the neural surrogate.

Theorem ((d+1)-Burgers' PDE : An Upper Bound for the L^2−Risk)

Let $\boldsymbol{u} \in C^1(D \times [t_0, T])$ be the unique solution of the $(d+1)-$dimensional Burgers' PDE. Then for any C^1 surrogate solution, say $\boldsymbol{u}_\theta$, the L^2-risk with respect to the true solution is bounded as,

$$\log\left(\int_\Omega \|\boldsymbol{u}(\boldsymbol{x}, t) - \boldsymbol{u}_\theta(\boldsymbol{x}, t)\|_2^2 \, \mathrm{d}\boldsymbol{x}\, \mathrm{d}t\right) \leq \log\left(\frac{C_1 C_2}{4}\right) + \frac{C_1}{\sqrt{2}} \qquad (1)$$

where,

$$C_1 = d^2 \|\nabla \boldsymbol{u}_\theta\|_{L^\infty(\Omega)} + 1 + d^2 \|\nabla \boldsymbol{u}\|_{L^\infty(\Omega)}$$

$$C_2 = \int_D \|\mathcal{R}_t\|_2^2 \, \mathrm{d}\boldsymbol{x} + \int_\Omega \|\mathcal{R}_{pde}\|_2^2 \, \mathrm{d}\boldsymbol{x}\, \mathrm{d}t + d^2 \|\nabla \boldsymbol{u}_\theta\|_{L^\infty(\Omega)} \int_\Omega \|\boldsymbol{u}_\theta\|_2^2 \, \mathrm{d}\boldsymbol{x}\, \mathrm{d}t + d^2 \|\nabla \boldsymbol{u}\|_{L^\infty(\Omega)} \int_\Omega \|\boldsymbol{u}\|_2^2 \, \mathrm{d}\boldsymbol{x}\, \mathrm{d}t$$

Some key points about this bound

- The RHS, the bound on the true risk is computable without knowing the exact true solution - and it can be estimated just by knowing an upper bound on $\|\nabla \boldsymbol{u}\|_{L^\infty(\Omega)}$ and $\|\boldsymbol{u}\|_2$.
- The proof of the above bound does not use $\nabla \cdot \boldsymbol{u} = 0$ condition on the true solution - and it allows for flexible boundary conditions. These features ensure that this bound can be used for known blow-up solutions for this PDE!

(1+1)-Burgers' PDE

We analyse the following PDE in the domain $x \in [-1, 1]$ and $t \in [-1 + \delta, \delta)$

$$u_t + u u_x = 0\ ; \quad u(x, -1 + \delta) = \frac{x}{-2 + \delta}$$

$$u(-1, t) = \frac{1}{1 - t}\ ; \quad u(1, t) = \frac{1}{t - 1}$$

Then for the neural net $\boldsymbol{u}_\theta(\boldsymbol{x}, t) \in \mathbb{R}$, we define the residuals as

$$R_{int,\theta}(x, t) := \partial_t u_\theta(x, t) + u_\theta(x, t) \cdot \partial_x u_\theta(x, t)$$

$$R_{tb,\theta}(x) := u_\theta(x, -1 + \delta) - \frac{x}{-2 + \delta}$$

$$R_{sb,-1,\theta}(t) := u_\theta(-1, t) - \frac{1}{1 - t}\ ; \quad R_{sb,1,\theta}(t) := u_\theta(1, t) - \frac{1}{t - 1}$$

Theorem ((1+1)-Burgers' PDE : An Upper Bound for the L^2−Risk)

Let $u \in C^1((-1 + \delta, \delta) \times (-1, 1))$ be the unique solution of the (1+1)-D Burgers' equation for any $k \geq 1$ and $u^* = u_{\theta^*}$ be the neural surrogate. Then the population risk of it is bounded as,

$$\left(\int_{-1+\delta}^{\delta} \int_{-1}^{1} |u(x, t) - u_\theta(x, t)|^2 \, dxdt\right)^{\frac{1}{2}}$$
$$\leq \left[1 + Ce^C\right] \left[\int_{-1}^{1} \mathcal{R}_{tb,\theta^*}(x) dx + \int_{-1+\delta}^{\delta} \int_{-1}^{1} \mathcal{R}_{int,\theta^*}^2(x, t) dxdt\right.$$
$$+ 2C_{2b} \left(\int_{-1+\delta}^{\delta} \mathcal{R}_{sb,-1,\theta^*}^2(t) dt + \int_{-1+\delta}^{\delta} \mathcal{R}_{sb,1,\theta^*}^2(t) dt\right)$$
$$\left. + 2C_{1b} \left(\left(\int_{-1+\delta}^{\delta} \mathcal{R}_{sb,-1,\theta^*}^2(t) dt\right)^{\frac{1}{2}} + \left(\int_{-1+\delta}^{\delta} \mathcal{R}_{sb,1,\theta^*}^2(t) dt\right)^{\frac{1}{2}}\right)\right] \qquad (2)$$

Some key points about this bound

- An interesting thing to note here is that, even though we set up the initial and boundary conditions to reflect a finite-time singularity, our bound for the L_2 population risk indicates the stability of the PINN risk, defined as,

$$\mathbb{E}[|\mathcal{R}_{int,\theta}(x, t)|^2] + \mathbb{E}[|\mathcal{R}_{tb,\theta}|^2] + \mathbb{E}[|\mathcal{R}_{sb,-1,\theta}|^2] + \mathbb{E}[|\mathcal{R}_{sb,1,\theta}|^2]$$

- The "stability" here is in the precise sense (C. Wang et al. 2022), that if this PINN risk is found to be $\mathcal{O}(\varepsilon)$ then that would directly imply that the L_2 population risk is also $\mathcal{O}(\varepsilon)$.

(1+1)-Burgers' PDE : Plots

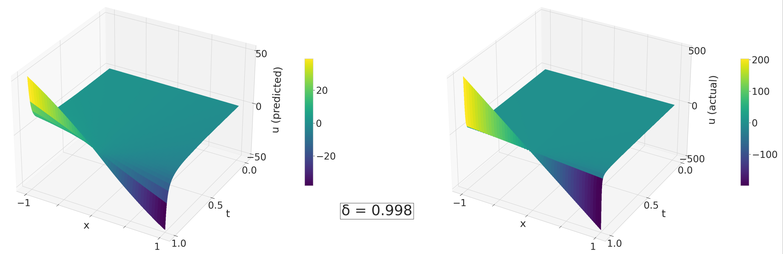
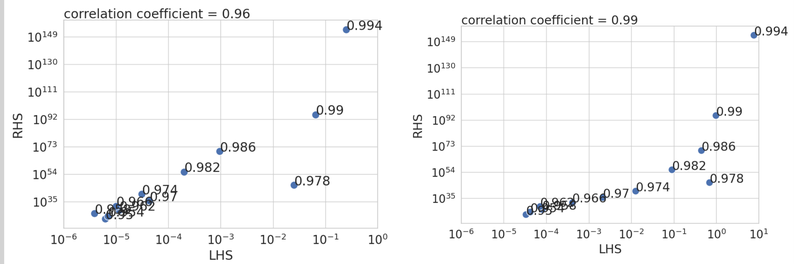


Figure: A demonstration of the visual resemblance between the neurally derived solution (left) and the true solution (right), when the max time is $\delta = 0.998$ (blow-up at $t = 1$)

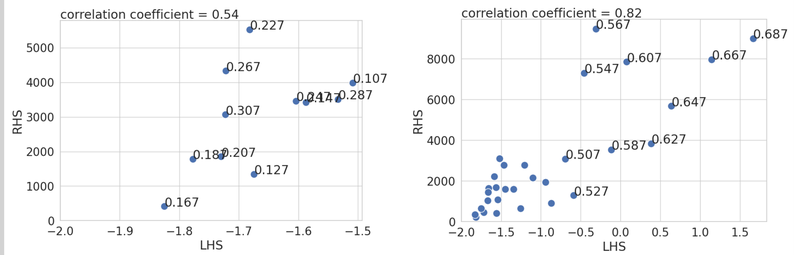


(a) width=30 (b) width=300

Figure: This figure shows the RHS vs LHS plot of Equation (2) for different values of δ for PINNs of 2 different widths. (Recall that here the blow up is at $\delta = 1$)

We note that the Pearson correlation coefficient between the LHS and the RHS increases when the width of the neural net.

(2+1)-Burgers' PDE : LHS vs RHS Plots



(a) width=30 (b) width=100

Figure: This figure shows the RHS vs LHS plot of Equation (1) from Theorem 1 for different values of δ for PINNs of 2 different widths. (Recall that here the blow up is at $\delta \sim 0.7$)

References

Wang, Chuwei et al. (2022). "Is L^2 Physics Informed Loss Always Suitable for Training Physics Informed Neural Network?" In: *Advances in Neural Information Processing Systems* 35, pp. 8278–8290.

Wang, Yongji et al. (2022). *Asymptotic self-similar blow up profile for 3-D Euler via physics-informed neural networks.* arXiv: 2201.06780 [math.AP].